

**2024 24th International Conference on Control, Automation and Systems (ICCAS)**

Oct 29 - Nov 1, 2024, Jeju Shinhwa World, Jeju, Korea

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Last updated on October 15, 2024. This conference program is tentative and subject to change

Technical Program for Wednesday October 30, 2024

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WeAT2	Room T2
<u>Award Session 1</u>	Oral Session
Chair: Ahn, Kyoung Kwan	University of Ulsan
09:00-09:15, Paper WeAT2.1	
<i>3D LIDAR SLAM and Map Based Illegal Parking Detection for Electric Scooter</i>	
Ha, ChangWan	Jeonbuk National University
Kim, KangGeon	Jeonbuk National University
Jang, TaeHwan	Jeonbuk National University
Chae, GyoungTae	Jeonbuk National University
Jo, HyungGi	Jeonbuk National University
09:15-09:30, Paper WeAT2.2	
<i>Development of Neutral Zone-Based Lumbar Support Robotic Device Utilizing an Electroadhesive Clutch</i>	
Lee, Garam	Seoul National University
Sung, Eunho	Seoul National University
Kim, Joowan	Seoul National University Hospital
Park, Jaeheung	Seoul National University
Kim, Keewon	Seoul National University Hospital
09:30-09:45, Paper WeAT2.3	
<i>CSAC Drift Modeling Considering GPS Signal Quality in the Case of GPS Signal Unavailability</i>	
Park, Seunghyeon	Yonsei University
Rhee, Joon Hyo	Korea Research Institute of Standards and Science
09:45-10:00, Paper WeAT2.4	
<i>Enhancing the Workspace of a Magnetic Soft Robot with a Movable Opposing Magnet</i>	
Park, Joo-Won	DGIST
Park, Sukho	DGIST
10:00-10:15, Paper WeAT2.5	
<i>Life Prediction and Life Extension Control of Multi-Motor Synchronous Control Systems</i>	
Yan, Weijun	Northwestern Polytechnical University
Liu, Zhenbao	Northwestern Polytechnical University
Jia, Zhen	Northwestern Polytechnical University
WeAT3	Room T3
<u>Industrial Applications of Control 1</u>	Oral Session
Chair: Suh, Young Soo	Univ. of Ulsan
09:00-09:15, Paper WeAT3.1	
<i>High-Precision Position Control of Electro-Hydrostatic Actuator under System Uncertainties</i>	
Ding, Guyue	Tongji University
Bian, Yongming	Tongji University
Yuan, Zhicheng	Tongji University
Geimer, Marcus	Karlsruhe Institute of Technology
Helian, Bobo	Karlsruhe Institute of Technology
Chen, Li	Tongji University
09:15-09:30, Paper WeAT3.2	
<i>Trajectory Tracking Control of Tracked Vehicle with Consideration of Track Slip and Disturbance</i>	
Huynh, Thinh	Pukyong National University
Nguyen, Van-Trang	Ho Chi Minh City University of Technology and Education
Kim, Young-Bok	Pukyong National University
09:30-09:45, Paper WeAT3.3	
<i>Excavation Trajectory Planning Method Based on Path Coarse Interpolation</i>	

Wang, Ziwei	Tongji University
Liu, Guangjun	Tongji University
Helian, Bobo	Karlsruhe Institute of Technology
Wang, Yunfei	Shanghai Engineering Research Center for Safety Intelligent Cont
09:45-10:00, Paper WeAT3.4	
<i>Stiffness Recognition Method Based on Modal Vibration for Serial Robotic Systems</i>	
Liang, Siyu	Tongji University
Liu, Guangjun	Tongji University
Helian, Bobo	Karlsruhe Institute of Technology
Wang, Yunfei	Shanghai Engineering Research Center for Safety Intelligent Cont
10:00-10:15, Paper WeAT3.5	
<i>Anomaly Detection System for Semiconductor Deposition Process Using Transformer</i>	
Ha, Min-Ho	Chungbuk National University
Arshad, Saba	Chungbuk National University
Chae, Min-Seon	Chungbuk National University
Yun, Sung-Chul	Chungbuk National University
Kim, Seung-Hwan	GSF Solution
Park, Tae-Hyoung	Chungbuk National University
10:15-10:30, Paper WeAT3.6	
<i>Koopman Fault-Tolerant Model Predictive Control for Multirotor with Actuator Fault</i>	
Lee, Jayden Dongwoo	KAIST
Kim, Hyeonmin	KAIST
Kim, Yoonseong	Univ. Dongguk
Bang, Hyechoong	KAIST
WeAT4 Room T4	
Autonomous Vehicle Systems 1 Oral Session	
Chair: Morita, Yoshifumi	Nagoya Institute of Technology
09:00-09:15, Paper WeAT4.1	
<i>Metaverse Aided Teleoperation Scene Prediction Optimization Using Depth Camera Transformation Based on Relative Velocities</i>	
An, Jonghyeok	Korea National University of Science and Technology
Jeong, Chanyeong	Gwangju Institute of Science and Technology
Cha, Hyunrok	Korea Institute of Industrial Technology
Hwang, Myeonghwan	Korea Institute of Industrial Technology
Yoon, Seunggha	Korea Institute of Industrial Technology
Kim, Eugene	Korea Institute of Industrial Technology
09:15-09:30, Paper WeAT4.2	
<i>A Guided Control Approach for Autonomous Docking Considering Estimation and Motion Capabilities</i>	
Im, Jinho	Keimyung University
Hong, Seonghun	Keimyung University
09:30-09:45, Paper WeAT4.3	
<i>Novel Unified Control Strategy for a Compound VTOL Vehicle</i>	
Chen, Lulu	Northwestern Polytechnical University
Liu, Zhenbao	Northwestern Polytechnical University
Dang, Qingqing	Northwestern Polytechnical University
Zhao, Wen	Northwestern Polytechnical University
09:45-10:00, Paper WeAT4.4	
<i>Pure Pursuit Method with Variable Ratio of Front and Rear-Wheel Steering Angles for Four-Wheel Independent Steering Vehicles</i>	
Nogami, Narumi	Nagoya Institute of Technology
Ozu, Yuma	Nagoya Institute of Technology
Morita, Yoshifumi	Nagoya Institute of Technology
Suzuki, Yuta	DENSO CORPORATION
Takahashi, Tomoya	DENSO CORPORATION
10:00-10:15, Paper WeAT4.5	
<i>Enhancing Vehicle Safety in High-Traffic Environments: An Optimal Segmented Trajectory Planning Approach</i>	
Kim, Dongryul	Kyungpook National University
Han, Kyoungseok	Hanyang University
WeAT5 Room T5	
Biomechanics and Biomedical Systems Oral Session	
Chair: Park, Sukho	DGIST
09:00-09:15, Paper WeAT5.1	

System for Analysis and Verification of Exercise Postures with Equipment

Anuntachai_Anuntapatt	KMITL
Kamcharoen_Chayane	KMITL
Boriboon_Prakasit	KMITL

09:15-09:30, Paper WeAT5.2

Magnetically Actuated Capsule with Multiple Sampling Capability for Gut Microbiome Research

Park_Sanghyeon	Daegu Gyeongbuk Institute of Science & Technology
Park_Sukho	DGIST

09:30-09:45, Paper WeAT5.3

Soft Manipulator for Minimally Invasive Surgery

Asim_Muhammad	Department of Mechanical Engineering, Institute of Space Technol
Turab_Huzeifa	Department of Mechanical Engineering Institute of Space Technolo
Ghafoor_Ulman	Pusan National University
Qureshi_Yumna	Department of Mechanical Engineering Institute of Space Technolo
Ikram_Rugia	Department of Mechanical Engineering Institute of Space Technolo
Bhutta_Muhammad_Baheel	Department of Electrical and Computer Engineering, University Of

09:45-10:00, Paper WeAT5.4

Intraoperative Tumor Boundary Detection Using Sweeping Palpation in Robot-Assisted Minimally Invasive Surgery (RMIS)

Hong_Jeongbin	DGIST
Park_Sukho	DGIST

10:00-10:15, Paper WeAT5.5

Enabling Frail Subjects Telemonitoring: A Sensor Network System for Enhanced Health Assessment

Bodo_Giulia	Politecnico Di Torino & Istituto Italiano Di Tecnologia
Sciortino_Giulio	Istituto Italiano Di Tecnologia
PETRUSHIN_Alexey	Fondazione Istituto Italiano Di Tecnologia
Seminario_Emanuele	Geriatrics Unit, Department of Geriatric Care, Neurology and Reh
Morganti_Wanda	Geriatrics Unit, Department of Geriatric Care, Neurology and Reh
Baillo_Gian_Luca	Fondazione Istituto Italiano Di Tecnologia
Beltran-Gonzalez_Carlos	Istituto Italiano Di Tecnologia (IIT)
Pilotto_Alberto	Galliera Hospital
Laffranchi_Matteo	Istituto Italiano Di Tecnologia
Del_Bug_Alessio	Istituto Italiano Di Tecnologia

WeAT6	Room T6
Daejeon-Chungcheong Dissemination Session	Oral Session

Chair: Kim_Min_Jun	KAIST
Organizer: Kim_Min_Jun	KAIST

09:00-09:15, Paper WeAT6.1

Low Clearance Hinge Joint Mechanism Based on 3D Printing on Sheet Fabrication Methodology (I)

Jang_Jaehyung	Korea Advanced Institute of Science and Technology
Shin_Fuibin	Korea Advanced Institute of Science and Technology
Okamura_Allison_M.	Stanford University
Ryu_Jee-Hwan	Korea Advanced Institute of Science and Technology

09:15-09:30, Paper WeAT6.2

An Integrated Framework for Autonomous Navigation of Legged Robots in Urban Environments (I)

Kim_Sangmin	Korea Advanced Institute of Science and Technology
Park_Jaehyun	Korea Advanced Institute of Science and Technology
Kong_Jei	Korea Advanced Institute of Science and Technology
Park_Hae-Won	Korea Advanced Institute of Science and Technology

09:30-09:45, Paper WeAT6.3

Design of Hospital Bed Mover Mobile Robot (I)

Shin_Heechul	Korea University, Sejong
Cheong_Joono	Korea University

WeAT7	Room T7
Deep Learning and Machine Vision Applications 1	Oral Session

Chair: Kim_Sungho	Yeungnam University
Organizer: Kim_Sungho	Yeungnam University
Organizer: Hwang_Youngbae	Chungbuk National University
Organizer: Lee_Wangheon	Hansei University

09:00-09:15, Paper WeAT7.1

Soiling Builds up Gradually : Soiling Semantic Segmentation (I)

Kim_Taewoo	Chungbuk National University
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Lee, Wangheon	Hansei University
Hwang, Youngbae	Chungbuk National University
09:15-09:30, Paper WeAT7.2	
RetinexPulse: Tough Remote Optical Volume Pulse Wave Measurement Technique in Various Illumination Environments (I)	
Lee, Seongryeong	Yeungnam University
Kim, Sungho	Yeungnam University
09:30-09:45, Paper WeAT7.3	
Addressing Relative Pose Impact on UWB Localization: Dataset Introduction and Analysis (I)	
choe, junhyeok	Inha University
Shim, Inwook	Inha University
09:45-10:00, Paper WeAT7.4	
Compact Monocular Camera-Based Indoor Object Trajectory Extraction System for Enhanced Monitoring (I)	
Kim, Donyung	Yeungnam University
Kim, Sungho	Yeungnam University
10:00-10:15, Paper WeAT7.5	
Estimating Emissivity Based on the Difference between Indoor Background Temperature and Target Temperature (I)	
Jang, Yewon	Yeungnam Univ
Kim, Sungho	Yeungnam University
WeAT8	Room T8
Multi-Agent Systems	Oral Session
Chair: Nguyen, Khanh Hieu	Sejong University
09:00-09:15, Paper WeAT8.1	
Decentralized Multi-Agent Reinforcement Learning Exploration with Inter-Agent Communication-Based Action Space	
Calzolari, Gabriele	Luleå University of Technology
Sumathy, Vidya	Luleå University of Technology
Kanellakis, Christoforos	Luleå University of Technology
Nikolakopoulos, George	Luleå University of Technology
09:15-09:30, Paper WeAT8.2	
Multi-Agent Network Randomization Method for Robust Knowledge Transfer in Deep Multi-Agent Reinforcement Learning	
Kim, Dohyun	Agency for Defense Development
Jung, Hoseong	Agency for Defense Development
Bae, Jung Ho	Agency for Defense Development
09:30-09:45, Paper WeAT8.3	
An Leader-Following Consensus of Multi-Agent System Via Centrality Analysis	
Lee, ChanWoo	Chungbuk National University
Lee, YongGwon	Chungbuk National University
Kim, Yeongjae	Chungbuk National University
Lee, Seunghoon	National Korea Maritime and Ocean University
Kwon, OhMin	Chungbuk National University
09:45-10:00, Paper WeAT8.4	
Numerical and Experimental Assessment of Distributed Model Predictive Control of Multi-Agent Circular Formation	
Kim, Seongheon	Gyeongsang National University
Araujo Pimentel, Guilherme	University of Mons
Kim, Yoonsoo	Gyeongsang National University
Vande Wouwer, Alain	University of Mons
10:00-10:15, Paper WeAT8.5	
Improved Sampled-Data Consensus Control for Multi-Agent Systems with Time Delay	
Nguyen, Khanh Hieu	Sejong University
Kim, Sung Hyun	University of Ulsan
WeP1S	Room T11
Plenary Lecuture 1: Data-Driven Control Based on Behavioral Systems Theory	Plenary Lecture
Chair: Eun, Yongsoo	DGIST
WeBT2	Room T2
Koh Young AI Competition	Oral Session

WeBT3	Room T3
<u>Industrial Applications of Control 2</u>	Oral Session
Chair: Lee, Youngwoo	Hanyang University
13:00-13:15, Paper WeBT3.1	
<u>Minimum Backflow Power and ZVS Condition for a Dual Active Bridge DC/DC Converter with Extended Phase Shift Modulation Based on KKT Conditions</u>	
Nguyen, Ngoc Nam	Phenikaa University
Nguyen, Ngoc-Duc	Seoul National University of Science and Technology
Lee, Young il	Seoul National University of Science and Technology
13:15-13:30, Paper WeBT3.2	
<u>PI Controller for Electric Vehicle Charger Using Deep Reinforcement Learning</u>	
Park, Keunhoon	Hanyang University
Kim, Seojun	Hanyang University
Lee, Youngwoo	Hanyang University
13:30-13:45, Paper WeBT3.3	
<u>Modeling and Charging Method for Supercapacitor Energy Management Systems</u>	
Kim, Seojun	Hanyang University
Jang, Taeseung	Hanyang University
Lee, Youngwoo	Hanyang University
13:45-14:00, Paper WeBT3.4	
<u>Evaluation of Hardware-In-The-Loop-Based Converter/Inverter Digital Twins for Microgrids Using the OPC Unified Architecture Platform</u>	
LEE, CHANGDAE	Seoul National University of Science and Technology
Nguyen, Ngoc-Duc	Seoul National University of Science and Technology
Lee, Dong Hun	Seoul National University of Science and Technology
Lee, Young il	Seoul National University of Science and Technology
14:00-14:15, Paper WeBT3.5	
<u>A Real-Time Simulation to Evaluate the Peak Shaving Energy Management System for Vanadium Redox Flow Battery-Based Electric Vehicle Charging Station</u>	
Nguyen, Ngoc-Duc	Seoul National University of Science and Technology
LEE, CHANGDAE	Seoul National University of Science and Technology
Lee, Young il	Seoul National University of Science and Technology
14:15-14:30, Paper WeBT3.6	
<u>Nonlinear Cascade Fuzzy Control of a Water Droplet Generator in an Subsonic Icing Wind Tunnel</u>	
Hernández-Hernández, César	ESIGELEC/IRSEEM
Chevet, Thomas	ESIGELEC/IRSEEM
Thabet, Rihab El Houda	ESIGELEC/IRSEEM
Sirroulomb, Vincent	ESIGELEC/IRSEEM
Langlois, Nicolas	ESIGELEC/IRSEEM
WeBT4	Room T4
<u>Autonomous Vehicle Systems 2</u>	Oral Session
Chair: Lee, Seong-Min	Jeju National University
13:00-13:15, Paper WeBT4.1	
<u>Vehicle Speed Prediction Model Using Neural Network for Model Predictive Control Depending on Future Prediction</u>	
Ishibashi, Ryo	Tokyo Denki University
Hidaka, Koichi	Tokyo Denki University
13:15-13:30, Paper WeBT4.2	
<u>Digital Twin Simulation of Teleoperation: Communication Delay Compensation Using LSTM Network</u>	
Kang, Hwanhee	Korea National University of Science and Technology
Cha, Hyunrok	Korea Institute of Industrial Technology
Hwang, Myeonghwan	Korea Institute of Industrial Technology
Yoon, Seunggha	Korea Institute of Industrial Technology
Kim, Eugene	Korea Institute of Industrial Technology
13:30-13:45, Paper WeBT4.3	
<u>Discrete-Time (O, S, B)-Alpha-Dissipative Active Braking Controller Synthesis Considering Articulation Angle of Bus-Trailer Systems</u>	
Kang, Rae Cheong	Korea University
Ahn, Woo Jin	Korea University
Lee, Yong Jun	Korea University
Lim, Myo-Taeg	Korea University
13:45-14:00, Paper WeBT4.4	
<u>Innovative Powertrain for Hybrid Hydraulic Excavators</u>	

NGUYEN_HIEN	University of Ulsan
Kwak_KyungShin	University of Ulsan
Ahn_KyoungKwan	University of Ulsan
14:00-14:15, Paper WeBT4.5	
Optimal Speed and Charging Planner for Electric Vehicles: MINLP, Relaxed NLP, and MIQP Approaches	
Gwon_Minwoo	Inha University
kim_jiwon	Inha University
Yoo_Seungjun	Inha University
Kim_Kwangki	Inha University
14:15-14:30, Paper WeBT4.6	
Experimental Comparison of Tightly and Loosely Coupled Kalman Filters for 6D Pose Estimation Using UWB and IMU Sensors	
Nam_Seungwon	Inha University
Kim_Yonghee	Inha University
Sim_Hyunjae	Inha University
Kim_Kwangki	Inha University
WeBT5	Room T5
Medical and Rehabilitation Robot	Oral Session
Chair: Kang_Brian Byunghyun	Sejong University
13:00-13:15, Paper WeBT5.1	
Vision-Based Intelligent Gait Environment Detection Algorithm for Assisting Stroke Patients with a Soft Ankle Exosuit	
Kang_Brian Byunghyun	Sejong University
Yang_Gayoung	Sejong University
13:15-13:30, Paper WeBT5.2	
Intelligent Soft Wearable Lumbar Support Robot System for Flat Back Syndrome	
Kang_Brian Byunghyun	Sejong University
Yang_Seungwon	Sejong University
Yang_Gayoung	Sejong University
Kim_Daeun	Sejong University
13:30-13:45, Paper WeBT5.3	
Fall Prediction for Self-Balancing Lower-Limb Exoskeletons Using Gaussian Process Classification	
Shahrokhshahi_Ahmadreza	Simon Fraser University
Arzanpour_Siamak	Simon Fraser University
Park_Edward J.	Simon Fraser University
13:45-14:00, Paper WeBT5.4	
Comparison of Muscle Activity During Active and Passive Training with the "iPARKO" Finger Extensor Facilitation Training Device	
Ishigaki_Shota	Nagoya Institute of Technology
Yokoyama_Reika	Nagoya Institute of Technology
Morita_Yoshifumi	Nagoya Institute of Technology
Tanabe_Hirofumi	Shonan University of Medical Sciences
14:00-14:15, Paper WeBT5.5	
Development and Preliminary Evaluation of a Real-Time Multiple Assistance System for Asymmetric Gait Rehabilitation Using an Instrumented Treadmill	
Park_Seoung Hoon	University of Houston
Lee_Beom-Chan	University of Houston
14:15-14:30, Paper WeBT5.6	
Gait Balance Improvement During Turning Gait through the Light Touch	
Kim_Minkyung	Gwangju Institute of Science and Technology
Lee_Junyeong	Gwangju Institute of Science and Technology
Kim_Hoyoung	GIST
Pyo_Sanghun	Gwangju Institute of Science and Technology
Yoon_Jungwon	Gwangju Institute of Science and Technology
WeBT6	Room T6
ECTI-ICROS Joint Organized Session on Advanced Control Designs and Applications	Oral Session
Chair: Suh_Young Soo	Univ. of Ulsan
Organizer: Jennawasin_Tanagorn	King Mongkut's University of Technology Thonburi
Organizer: Suh_Young Soo	Univ. of Ulsan
13:00-13:15, Paper WeBT6.1	

A Sparse Quadratic Programming Approach and Interior Point Method to Design Supervisory Model Predictive Control of Multi-Zone HVAC Systems (I)	
Sasananand, Pasitnat	Chulalongkorn University
Banjerdpongchai, David	Chulalongkorn University
13:15-13:30, Paper WeBT6.2	
Toothbrush Position Estimation Using Camera and Smart Toothbrush-Integrated Inertial Sensors (I)	
Pham, Thanh Tuan	University of Ulsan
Suh, Young Soo	Univ. of Ulsan
13:30-13:45, Paper WeBT6.3	
Stability Analysis of Delayed Neural Networks Via Modified Free-Matrix Based Integral Inequality (I)	
Park, Yongbeom	POSTECH
Lee, Hosub	POSTECH
PARK, POOGYEON	POSTECH
13:45-14:00, Paper WeBT6.4	
Disturbance-Observer Based PID Control Design for a Level Control System with Valve Stiction (I)	
Jennawasin, Tanagorn	King Mongkut's University of Technology Thonburi
WeBT7	Room T7
Deep Learning and Machine Vision Applications 2	Oral Session
Chair: Kim, Sungho	Yeungnam University
Organizer: Kim, Sungho	Yeungnam University
Organizer: Hwang, Youngbae	Chungbuk National University
Organizer: Lee, Wangheon	Hansei University
13:00-13:15, Paper WeBT7.1	
Real-Time High-Resolution SAR Imaging Using a Commercial W-Band Radar and Rail System (I)	
Kim, Minjun	Yeungnam University
Kim, Sungho	Yeungnam University
13:15-13:30, Paper WeBT7.2	
Non-Contact Respiratory Rate Estimation Using LWIR Camera (I)	
Kim, Jaeho	YeungNam University
Kim, Sungho	Yeungnam University
13:30-13:45, Paper WeBT7.3	
XAI-Driven Domain Gap Analysis for Improved SAR Image Classification (I)	
Song, Haekang	Yeungnam University
Kim, Sungho	Yeungnam University
13:45-14:00, Paper WeBT7.4	
Deep Learning-Based Calibration Method for an Augmented Reality Surgical Navigation System without Head-Mounted Optical Markers (I)	
Kim, Seoung kyung	Kyungpook National University
Kim, Min Young	Kyungpook National University
14:00-14:15, Paper WeBT7.5	
Fast Zernike Moment Computation Using PyTorch in a Multiple-GPU Environment (I)	
Suhng, ByuhngMunn	Univ. of Hansei
Lee, Wangheon	Univ. of Hansei
WeBT8	Room T8
China-Korea-Japan Joint Session on Advanced Control of Network and Dynamic Systems 1	Oral Session
Chair: Ahn, Hyo-Sung	Gwangju Institute of Science and Technology (GIST)
Organizer: Ahn, Hyo-Sung	Gwangju Institute of Science and Technology (GIST)
13:00-13:15, Paper WeBT8.1	
Stability Analysis of Impulsive Stochastic Networks with Deception Attacks and Input Delays (I)	
Yang, Ni	Harbin Institute of Technology
Su, Huan	Harbin Institute of Technology
PARK, POOGYEON	POSTECH
13:15-13:30, Paper WeBT8.2	
Hierarchical Control of CAV Using Adaptive CBF (I)	
Nakai, Yuta	Keio University
Namerikawa, Toru	Keio University

13:30-13:45, Paper WeBT8.3	
Design of Sparse Control with Minimax Concave Penalty (I)	
Hayashi_Naoki	Hiroshima University
Ikeda_Takuya	The University of Kitakyushu
Nagahara_Masaaki	Hiroshima University
13:45-14:00, Paper WeBT8.4	
Vision-Based Formation Control with Control Barrier Function (I)	
Won_Seung-Beom	Gwangju Institute of Science and Technology
Ahn_Hyo-Sung	Gwangju Institute of Science and Technology (GIST)
14:00-14:15, Paper WeBT8.5	
Consensus Speed of Multi-Agent Systems Driven by Simultaneously Fair Value Exchange (I)	
Sugiyama_Daiki	Kyoto University
Azuma_Shun-ichi	Kyoto University
WeP2S	Room T11
Plenary Lecuture 2: Positive Systems from the Modeling Viewpoint	Plenary Lecture
Chair: Suh_Young Soo	Univ. of Ulsan
WeCT3	Room T3
Control Applications 1	Oral Session
Chair: Han_Soohee	Pohang University of Science and Technology (POSTECH)
Organizer: Han_Soohee	Pohang University of Science and Technology (POSTECH)
16:10-16:25, Paper WeCT3.1	
2D Trajectory Optimization Using Spherical Pendulum Dynamic Constraints for Reducing Liquid Sloshing on Mobile Robots (I)	
Lim_Woo Joung	Pohang University of Science and Technology
Jung_Jooyeol	POSTECH(Pohang University of Science and Technology)
Han_Soohee	Pohang University of Science and Technology (POSTECH)
16:25-16:40, Paper WeCT3.2	
Faster GICP with Voxel-Based Computation: Dynamic Voxel Size Refinement for Enhanced Accuracy (I)	
Moon_Youngtae	Pohang University of Science and Technology
Hong_Junwoo	Pohang University of Science and Technology (POSTECH)
Park_Chaewon	POSTECH (Pohang University of Science and Technology)
Han_Soohee	Pohang University of Science and Technology (POSTECH)
16:40-16:55, Paper WeCT3.3	
Quadrotor Control Design with Spiking Neural Networks (I)	
Kim_Donghyeong	POSTECH(Pohang University of Science and Technology)
Lee_Changhyeon	POSTECH(Pohang University of Science and Technology)
Han_Soohee	Pohang University of Science and Technology (POSTECH)
16:55-17:10, Paper WeCT3.4	
Developing an Motion Blur-Aware OCR Pipeline for Industrial Environments (I)	
Park_Jonghyeok	Pusan National University
Cho_Jaehyung	Pohang University of Science and Technology (POSTECH)
Kim_Jangwon	Pohang University of Science and Technology (POSTECH)
Byu_Sanghyun	Pohang University of Science and Technology (POSTECH)
Jung_Jiseok	Pohang University of Science and Technology (POSTECH)
Lee_Jongsoo	Pohang University of Science and Technology (POSTECH)
Han_Soohee	Pohang University of Science and Technology (POSTECH)
Kim_Kyungjun	Pohang University of Science and Technology (POSTECH)
17:10-17:25, Paper WeCT3.5	
ROFT-VINS: Robust Feature Tracking-Based Visual-Inertial State Estimation for Harsh Environment (I)	
Park_Sanghyun	Pohang University of Science and Technology (POSTECH)
Han_Soohee	Pohang University of Science and Technology (POSTECH)
17:25-17:40, Paper WeCT3.6	
Enhanced Stability Criteria for the Synchronization of Neural Networks with Transmission Delays Using Sampled-Data Control (I)	
Moon_Seongrok	POSTECH
Na_Hyeon-Woo	POSTECH
PARK_POOGYEON	POSTECH
WeCT4	Room T4
SICE-ICROS Joint Organized Session : Robot Technology and Its Application	Oral Session

Chair: Hasegawa_Tadahiro	Shibaura Institute of Technology
Organizer: Hasegawa_Tadahiro	Shibaura Institute of Technology
Organizer: Jin_Sangrok	Pusan National University
16:10-16:25, Paper WeCT4.1	
<i>Competitive Swarm Optimizer with an Auxiliary Indication Function (I)</i>	
Kim_Jaehyung	Pusan National Univ
Kim_RyoonHan	Korea Institute of Machinery & Materials
Lee_Min Cheol	Pusan National University
16:25-16:40, Paper WeCT4.2	
<i>Training YOLOv8 Object Detection Model with Synthetic Images (I)</i>	
PARK_KYU	Pusan National University
Kim_Jaehyung	Pusan National Univ
Kim_RyoonHan	Korea Institute of Machinery & Materials
Kwon_Junghan	Pusan National University
Lee_Min Cheol	Pusan National University
16:40-16:55, Paper WeCT4.3	
<i>Enhancing Robot Action Efficiency in Clustered Environments through Pushing and Grasping Learning Utilizing Harris Corner Detection (I)</i>	
Woo_Daryeong	Pukyong National University
Kang_Jung-Ho	Pukyong National University
Lee_Kyung-Chang	Pukyong National University
16:55-17:10, Paper WeCT4.4	
<i>A Study of Dialect Identification Method Using Phoneme Texts (I)</i>	
Asafusa_Yohei	Shibaura Institute of Technology
Yoshimj_Takashi	Shibaura Institute of Technology
17:10-17:25, Paper WeCT4.5	
<i>A Study on Polishing Robot Status Recognition Using Sound Information (I)</i>	
KANDA_Kyosuke	Shibaura Institute of Technology
Yoshimj_Takashi	Shibaura Institute of Technology
17:25-17:40, Paper WeCT4.6	
<i>Seamless Switchable Self-Localization System between RTK-GNSS and NDT Scan Matching and Its Experimental Verification at Tsukuba Challenge 2023 (I)</i>	
Tanifuji_Naoto	Shibaura Institute of Technology
Hasegawa_Tadahiro	Shibaura Institute of Technology
Izutsu_Shunsuke	Shibaura Institute of Technology
Yuta_Shinichi	Shibaura Institute of Technology
WeCT5	Room T5
Localization and Mapping	Oral Session
Chair: Sim_Joo Yong	Sookmyung Women's University
16:10-16:25, Paper WeCT5.1	
<i>Efficient Graduated Non-Convexity for Pose Graph Optimization</i>	
Kang_Wonseok	Seoul National University
Kim_Jaehyun	Seoul National University
Chung_Jiseong	Seoul National University
Choi_Seungwon	Seoul National University
Kim-Taewan	Seoul National University
16:25-16:40, Paper WeCT5.2	
<i>Efficient Traversability Mapping for Enhanced Robotic Firefighting Operations</i>	
Choi_Young Joo	Korea Electronics Technology Institute
Park_Min Cheol	Korea Electronics Technology Institute
Lee_Han-Wool	Korea Electronics Technology Institute
Hwang_Jung-Hoon	Korea Eletronics Technology Institute
16:40-16:55, Paper WeCT5.3	

[Adaptive Calibration of Parameters for Accurate Vehicle Localization](#)

Lee_Yeina	Sookmyung Women's University
Kim_Jiyoung	Sookmyung Women's University
Lee_SeongBeon	Sookmyung Women's University
Kim_Kyoung_Eun	Sookmyung Women's University
Boin_Jeong	Sookmyung Women's University
lee_jina	Sookmyung Women's University
choi_dain	Sookmyung Women's University
Park_Jiwon	Sookmyung Women's University
Ryu_Yeonji	Sookmyung Women's University
Cho_Youngseo	Sookmyung Women's University
Park_Hyungchai	Stanford Center at the Incheon Global Campus
Sim_Joo_Yong	Sookmyung Women's University

16:55-17:10, Paper WeCT5.4

[Offload Scale Optimization Aided SLAM Using Reinforcement Learning for Digital Twin](#)

Byeon_Jaeguk	Korea National University of Science and Technology
Cha_Hyunrok	Korea Institute of Industrial Technology
Hwang_Myeonghwan	Korea Institute of Industrial Technology
Yoon_Seungba	Korea Institute of Industrial Technology
Kim_Eugene	Korea Institute of Industrial Technology

17:10-17:25, Paper WeCT5.5

[Performance Evaluation of Visual and LiDAR SLAM for Inland Waterways](#)

Filip_Lulian	German Aerospace Center (DLR)
Lass_Christoph	German Aerospace Center (DLR)
Medina_Daniel	German Aerospace Center (DLR)

17:25-17:40, Paper WeCT5.6

[Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR-\(Inertial\) SLAM](#)

Hahn_Sanghyun	Seoul National University
Oh_Seunghun	Seoul National University
Jung_Minwoo	Seoul National University
Kim_Ayoung	Seoul National University
Jung_Sangwoo	Seoul National University

17:40-17:55, Paper WeCT5.7

[A Study on the 4WS-Based Mobile Robot Driving Control Strategy for the Implementation of Outdoor Autonomous Driving](#)

Kim_Dohyun	Pukyong National University
Kim_Changwon	Pukyong National University

WeCT6

Room T6

Computer Vision

Oral Session

Chair: [Lee_SangJun](#) Jeonbuk National University

16:10-16:25, Paper WeCT6.1

[Revolutionizing Egg Quality Control: Advanced Prompt-Based Models for Automated Detection of Broken Eggs without the Need for Training](#)

Soontornnapan_Tomorn	King Mongkut's Institute of Technology Ladkrabang
Praneenattavee_Natdhanai	King Mongkut's Institute of Technology Ladkrabang
Ploysuwan_Tuchsanai	King Mongkut's Institute of Technology Ladkrabang

16:25-16:40, Paper WeCT6.2

[City-Scale Multi-Camera Tracking of Vehicles Using YOLOv9 and ByteTrack](#)

Nguyen_Huy-Hung	Sungkyunkwan University
Tran_Duong_Nguyen-Ngoc	Sungkyunkwan Univ
Pham_Long_Hoang	Sungkyunkwan Univ
Tran_Tai_Huu-Phuong	Sungkyunkwan University
Jeon_Jae_Wook	Sungkyunkwan Univ

16:40-16:55, Paper WeCT6.3

[Attention Guided Deep Neural Network for Inspection of Surface Defects on Steel Products](#)

Jeong_Seyoung	Jeonbuk National University
Song_Jimin	Jeonbuk National University
Lee_SangJun	Jeonbuk National University

16:55-17:10, Paper WeCT6.4

[Skill Level Recognition in Writing for Elementary School Students in Thailand Using Image Processing Technology](#)

Anuntachai_Anuntapat	KMITL
Jarusitratti_Nattawat	KMITL

[Laohakul, Krittee](#)

KMITL

17:10-17:25, Paper WeCT6.5

[Context-Aware Recognition of Elevator Buttons Using a Sequential Training Methodology](#)[GHOSH, ARPAN](#)

Sung Kyunkwan University

[Joo, Kyeong-Jin](#)

Sung Kyun Kwan University

[Galvis Giraldo, Gilberto](#)

Sungkyunkwan University

[Kuc, Tae-Yong](#)

Sungkyunkwan University

17:25-17:40, Paper WeCT6.6

[Improved 3D Photon Counting Imaging Using Singular Value Decomposition \(SVD\)](#)[SONG, SEUNGWOO](#)

Kyushu Institute of Technology

[YEO, GILSU](#)

Kyushu Institute of Technology

[Kim, Hyun-Woo](#)

Kyushu Institute of Technology

[Cho, Myungjin](#)

Hankyong National University

[Lee, Min-Chul](#)

Kyushu Institute of Technology

WeCT7

Room T7

[Sensor and Signal Processing 1](#)

Oral Session

Chair: [Yamashita, Atsushi](#)

The University of Tokyo

16:10-16:25, Paper WeCT7.1

[Control Strategy and Cumulative Damage Analysis for Biaxial Load Fatigue Test of Wind Turbine Blades](#)[Ma, Yi](#)

Tongji University

[zhou, aiguo](#)

Tongji University

[Shi, JinLei](#)

Tongji University

[Zhao, Shiwen](#)

Tongji University

[Dong, Tao](#)

Tongji University

[zhu, yutian](#)

Tongji University

16:25-16:40, Paper WeCT7.2

[Clustering of Hammering Sounds and Identification of Defect Clusters Based on Acoustic Energy Per Impact for Detection of Concrete Defects](#)[Shoda, koki](#)

Tokyo University

[Louhi Kasahara, Jun Younes](#)

The University of Tokyo

[Asama, Hajime](#)

The University of Tokyo

[An, Qi](#)

The University of Tokyo

[Yamashita, Atsushi](#)

The University of Tokyo

16:40-16:55, Paper WeCT7.3

[Bi-Axial Fatigue Test Method for Ultra-Large Wind Turbine Blade Based on Tug Fatigue Testing System](#)[Shi, JinLei](#)

Tongji University

[zhou, aiguo](#)

Tongji University

[Dong, Tao](#)

Tongji University

[Ma, Yi](#)

Tongji University

[Zhao, Shiwen](#)

Tongji University

16:55-17:10, Paper WeCT7.4

[Image Correction for Early Detection of Landslides Using Optical Flow](#)[Hamuro, Kaito](#)

Faculty of Advanced Science and Technology, Kumamoto University,

[Yamaguchi, Teruo](#)

Kumamoto University

WeCT8

Room T8

[China-Korea-Japan Joint Session on Advanced Control of Network and Dynamic Systems 2](#)

Oral Session

Chair: [Ahn, Hyo-Sung](#)

Gwangju Institute of Science and Technology (GIST)

Organizer: [Ahn, Hyo-Sung](#)

Gwangju Institute of Science and Technology (GIST)

16:10-16:25, Paper WeCT8.1

[Sim-To-Real Policy Transfer in Deep Reinforcement Learning \(I\)](#)[Jang, Yoonsu](#)

POSTECH

[Yoon, Seongwon](#)

POSCO Holdings

[Baek, Jongchan](#)

Electronics and Telecommunications Research Institute (ETRI)

[Han, Soohye](#)

Pohang University of Science and Technology (POSTECH)

16:25-16:40, Paper WeCT8.2

[Some Recent Results on Stochastic Linear-Quadratic Differential Games \(I\)](#)[Moon, Jun](#)

Hanyang University

16:40-16:55, Paper WeCT8.3

[On Controllability of Networked Systems with Control and Transmission Sampled Data \(I\).](#)

Yang, Zixuan	Shanghai Jiao Tong University
Wang, Lin	Shanghai Jiao Tong University

16:55-17:10, Paper WeCT8.4

[Modeling Multi-Agent Systems Dynamics Guaranteeing Convergence into an Invariant Set \(I\).](#)

Yoshinaka, Mizuki	Kyoto University
Sakurama, Kazunori	Tottori University

17:10-17:25, Paper WeCT8.5

[Heterogeneous UAV Task Allocation Using Shapley Value-Based Clustering \(I\).](#)

Lee, Jong-Won	KAIST
Oh, Yunseo	KAIST
Choi, Han-Lim	KAIST